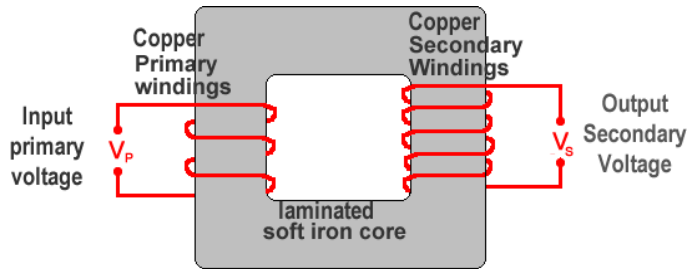


6(a)



The primary coil is connected to an alternating current which produces a continuously changing magnetic flux. This changing magnetic flux is linked to the secondary coil through the iron core.

Since there is continuously changing magnetic flux linkage to the secondary coil, by Faraday's law, an emf is induced in the secondary coil.

(b)(i) Frequency of voltage = 2 x frequency of power

$$T = 63/2 = 31.5 \text{ ms}$$

$$f = 1/T = 1/(31.5 \times 10^{-3}) = 31.7 \text{ Hz}$$

(ii) $P_{\max} = 17 \text{ W}$ (Read from graph)

(iii) $\langle P \rangle = P_{\max}/2 = 17/2 = 8.5 \text{ W}$

(iv)

$$\langle P \rangle = \frac{V_{\text{rms}}^2}{R}$$

$$8.5 = \frac{V_{\text{rms}}^2}{3.0}$$

$$V_{\text{rms}} = 5.0 \text{ V}$$

(v) $I_{\text{rms}} = V/R = 5.1/3.0 = 1.7 \text{ A}$